Note that quite often we will rewrite our market impact model using the variable *Size* as follows:

$$I^*_{bp} = a_1 \cdot Size^{a_2} \cdot \sigma^{a_3} \tag{5.3}$$

where,

$$Size = \frac{Shares}{ADV}$$

Test the Hypothesis

The fourth step in the scientific method consists of testing our hypothesis. But before we start testing the hypothesis with actual data it is essential to have a complete understanding of the model and dependencies across variables and parameters. We need to understand what results are considered feasible values and potential solutions.

Let us start with the interconnection between the a_1 and a_2 parameters. Suppose we have an order of 10% ADV for a stock with volatility of 25%. If we have $I^* = 84$ bp and $\hat{a}_3 = 0.75$, then any combination of a_1 and a_2 that satisfies the following relationship are potential feasible parameter values:

$$I^*_{bp} = a_1 \cdot Size^{a_2} \cdot \sigma^{0.75}$$

or,

$$84 \text{ bp} = a_1 \cdot (0.10)^{a_2} \cdot (0.25)^{0.75}$$

Solving for a_2 in the above formula yields:

$$a_2 = ln\left(\frac{I^*}{a_1 \cdot \sigma^{a^3}}\right) \cdot \frac{1}{ln(Size)}$$

Using the data in our example we have:

$$a_2 = ln\left(\frac{84}{a_1 \cdot 0.25^{a_{0.75}}}\right) \cdot \frac{1}{ln(0.10)}$$

However, not all combinations of a_1 and a_2 are feasible solutions to the model. Having prior insight into what constitutes these feasible values will assist dramatically when estimating and testing the model parameters. For example, we know that $a_2 > 0$ otherwise costs would be decreasing with order size and it would be less expensive to transact larger orders than smaller orders and there would be no need to slice and order and trade over time (not to mention an arbitrage opportunity).